W-F-421a

MARCH 7, 1940

SUPERSEDING
Fed. Spec. W-F-421
September 27, 1935

FEDERAL STANDARD STOCK CATALOG

Section IV

(Part 5)

FEDERAL SPECIFICATION

FOR

FLASHLIGHTS; ELECTRIC, HAND (WITHOUT BATTERIES)

This specification was approved on the above date by the Director of Procurement, for the use of all departments and establishments of the Government, and shall become effective not later than November 15, 1940. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE SPECIFICATIONS.

A-1. The following Federal Specifications, of the issue in effect on date of invitation for bids, shall form a part of this specification:

W–B 01—Batteries and Cells; dry.
W–L–111—Lamps; electric, incandescent, miniature, tungsten-filament—with corresponding Annual Supplement to W–L–111.

B. TYPES AND STYLES.

B-1. This specification covers portable hand lamps of tubular form commonly known as flashlights and equipped with lamps. Lantern types of flashlights are not covered by this specification. Dry cells and incandescent lamps as such are not covered. See current issues of W–B–101 and W–L–111, respectively.

B-1a. *Types.*—Two types of flashlights are covered by this specification: Type I consists of flashlights with seamless hard-drawn brass cases; type II consists of flashlights with rigid cases having a non-conducting exterior surface.

B-1b. *Styles.*—Type I flashlights consist of 11 styles, as shown in table I. Type II flashlights consist of 6 styles as shown in table I.

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TABLE I.—*Types and styles of flashlights*

Style description	Identifying code
Type I. Flashlights with brass cases:	
1. Nonfocusing, 2-cell	N-2-D.
2. Nonfocusing, 3-cell	N-3-D.
3. Spotlight, 2-cell	F-2-D.
4. Spotlight, 3-cell	F-3-D.
5. Searchlight, 3-cell	S-3-D.
6. Searchlight, 5-cell	S-5-D.
7. Right-angle, 2-cell	RF-2-D
8. Prefocused spotlight, 2-cell	PF-2-D.
9. Prefocused spotlight, 3-cell	PF-3-D.
10. Prefocused searchlight, 3-cell	PS-3-D.
11. Prefocused right-angle, 2-cell	PRF-2-D
Type II. Flashlights with nonconducting cases:	
12. Nonfocusing, 2-cell	NI-2-D.
13. Nonfocusing, 3-cell	NI-3-D.
14. Spotlight, 2-cell	FI-2-D.
15. Spotlight, 3-cell	FI-3D.
16. Prefocused spotlight, 2-cell	PFI-2-D.
17. Prefocused spotlight, 3-cell	PFI-3-D.

B–1c. *Definition of styles.*—

B–1c (1). Nonfocusing flashlights provide a diffused light projected from a head only slightly larger in diameter than the tube or body of the flashlight. These styles are designated by the letter "N" in the code.

Note.—Diffused light is light projected at a relatively wide angle and is useful only at relatively short distances.

B–1c (2). Spotlight flashlights provide a concentrated beam of light, projected from a head 2 to 2½ inches over-all in diameter. These styles are designated by the letter "F" in the code.

Note.—Concentrated light as referred to in these definitions is light projected in a nearly parallel beam, and is useful for illuminating objects at considerable distances.

B–1c (3). Searchlight flashlights provide a concentrated beam of light, projected from a head from 3 to 3¾ inches over-all in diameter. These styles are designated by the letter "S" in the code.

B–1c (4). Right-angle flashlights are similar in character to the spotlights but have the head arranged to project a concentrated beam at right angles to the body of the flashlight. These styles are designated by the letters "RF" in the code.

B–1c (5). Prefocused flashlights project a concentrated beam of light from a reflector provided with a lamp whose filament is fixed at the focal point of the reflector, and require no manual adjustment to focus the beam. In the code designation for the prefocused lights the letter "P" is prefixed for the style to which the prefocused feature is applied.

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B-1d. *Identifying code.*—A code consisting of three terms for convenient designation of styles is given in table I. The three terms are:

1. A letter or letters designating the kind of light projection and kind of head.
2. A numeral showing the number of dry cells required to operate the flashlight.
3. A letter designating the size of dry cell required for the flashlight. (See Federal Specification W-B-101).

B-1e. *Definitions of types.*—

B-1e (1). Type I flashlights are highest quality for general purpose use, and shall have exterior finish of full chromium plate or of black enamel with chromium-plated fittings, and with type 1 switch (as per paragraph E-3N). Style flashlights of this type shall have class 2 reflectors. Styles F, S, RF, PF, PS, and PRF flashlights of this type shall have class 1 reflectors and double thickness glass lenses (or plastic lenses if so specified).

B-1e (2). Type II flashlights are of highest quality, strong, durable, and externally insulated. They are designated by the letter "I" in the identifying code of table I. Such flashlights shall be of rigid construction, and the outside surface throughout shall be of an insulating or nonconducting material. They shall have type II switches (as per par. E-3b). Styles FI and PFI flashlights shall have class 1 reflectors. Style NI flashlights shall have class 2 reflectors. Lenses shall be of a suitable plastic (see sec. E-2b) or of glass as defined for type I flashlights.

Note.—Classes of reflectors and lenses are described in paragraphs E-1 and E-2.

C. MATERIAL AND WORKMANSHIP.

C-1. All flashlight cases shall be free from defects in materials and workmanship.

D. GENERAL REQUIREMENTS.

D-1. *Location of bulb in focusing (spotlight and searchlight) type flashlights.*—The lamp bulb shall be located so that the lamp filament will be at the focal point, or may be brought to the focal point, of the reflector.

D-2. Each flashlight case, excepting prefocused flashlights, shall be equipped with a minaiture tungsten-filament lamp (Federal Specification W-L-111 with corresponding annual supplement) as follows:

Two-cell flashlight cases, trade No. 14 lamp.
Three-cell flashlight cases, trade No. 13 lamp.
Five-cell flashlight cases, trade No. 31 lamp.

D-3. Pre-focused styles (PF-2-D, PF-3-D, PS-3-D, PRF-2-D, PFI-2-D and PFI-3-D) shall be equipped with "prefocused" miniature tungsten filament lamps. These lamps have a special flanged base designed to fit accurately into a recess in the back of the reflector.

Two-cell styles shall be provided with lamp, trade No. PR-2, unless otherwise specified.
Three-cell styles shall be provided with lamp, trade No. PR-3, unless otherwise specified.

E. DETAIL REQUIREMENTS.

E-1. *Reflectors.*—

E-1a. Flashlights with style N or NI heads may have reflectors whose shape is optional with the manufacturer. Flashlights having styles F, S, RF, PF, PS, PRF, FI, or PFI heads shall have parabolic reflectors. Reflectors shall be designated as class 1 or 2, as follows:

E-1b. Class 1 reflectors shall be of brass or die-cast metal, plated with silver. The reflecting surface shall be highly polished, and lacquered.

E-1c. Class 2 reflectors shall be the same as class 1 but need not be polished.

E-2. *Lenses.*—

E-2a. Lenses for flashlights having style N or NI heads shall be of the convex type.

E-2b. All glass lenses shall be clear and free of bubbles. Plastic lenses shall be clear, not readily subject to scratching in use, and not subject to distortion at 120° F., nor to shattering or breaking at 15° F. in ordinary use. Lenses shall be of such size and shall be so mounted as to permit full use of the reflector area.

E-2c. Double-thickness glass lenses shall be not less than 0.117 inch thick.

E-2d. Means shall be provided in focusing and prefocused styles of type I flashlights which will serve to prevent the lens from falling out of the lens ring when removed from the case to replace lamps. The lens shall be so located that the lamp bulb can not come into contact with it.

E-3. *Switches.*—

E-3a. Switches shall be conveniently located and securely attached to the flashlight. The finish shall be in keeping with and of a quality not inferior to the finish of the rest of the flashlight.

E-3b. Type I switches shall be provided with a "permanent-on" position and a "flashing" position. They shall be provided with means to prevent accidental closing of the circuit. Type II switches shall be so constructed that the entire exterior is of nonconducting material and be strong enough to resist accidental breakage at temperatures as low as 15° F. They shall resist accidental movement into the "on" position.

E-3c. Switches shall be well constructed of suitable materials and be capable of 25,000 cycles of normal operation without mechanical or electrical failure. For use in the tropics special materials may be specified.

E-4. *Body tube.*—

E-4a (1). The assembled tube shall freely receive a battery of maximum allowable diameter as specified in Federal Specification No. W-B-101 and the effective inside diameter shall not exceed 1.50 inches. It shall be ribbed or fluted lengthwise for at least three quarters of the length of the tube to improve grip and increase strength.

E-4a (2). Tubes for type I flashlights shall be of seamless hard-drawn brass, unannealed after the last forming operation, and expanded or reduced at the bottom end to provide strength and rigidity. Thickness shall not be less than as shown in table II.

TABLE II.—*Minimum wall thickness*

	Minimum wall thickness
	Inch
2-cell length tubes	0.010
3-cell length tubes	.011
5-cell length tubes	.012

E–4a (3). Tubes for type II flashlights shall be of a nonconducting insulating material such as vulcanized fiber, molded plastics, or rubber, reinforced on the inside with brass throughout or at least at the two ends. If vulcanized fiber or soft rubber is used, the tube shall be reinforced throughout its length with brass. The complete tube shall be so constructed and of such material that it shall have the following properties:

E–4a (3) a. The insulation of the tube when dry shall not break down when a potential of 1,000 volts, 60 cycles is applied for 3 minutes between any point on the outside of the tube to any point on the inside of the tube.

E–4a (3) b. The tube as a whole shall not change shape in ordinary use if continuously exposed to an external temperature of 120° F.

E–4a (3) c. The tube in a complete flashlight with cells, after having been cooled to a temperature of 15° F, shall not shatter or break if dropped three times through a distance of 3 feet to a concrete floor.

E–4a (3) d. The nonconducting material shall be resistant in ordinary use to the deteriorating effects of salt air, water, oils, greases, gasoline, alcohols, alkalies, and acids.

E–4b. Theaded parts shall be so fitted as to move relative to each other without difficulty but not so loosely fitted as to allow threads to jump or parts to change adjustment. In type II flashlights the threaded junction of the fittings with the tube shall, at least on one surface, be of brass to avoid the binding of the threads which is apt to occur if both male and female threads are of nonconducting material.

E–4c. There shall be no unobstructed opening to the inside of the flashlight as, for instance, through eyelets or hollow rivets.

E–4d. In type I flashlights means shall be provided, such as an octagonal lens ring, to prevent the flashlight from rolling freely.

E–4e. Type I flashlights shall be provided with a wire hanger fitted to the bottom cap in a manner to allow its being retained, when not in use, in an unobtrusive position.

E–4f. Cells shall be held in the case under sufficient pressure to insure good contact and to prevent breaking the circuit when the flashlight is shaken by the hand. Cells shall be readily replaceable.

E–4g. Styles F, FI, RF, and S flashlights shall be readily adjusted for correct focus of the light beam by a mechanism which will retain such adjustment under conditions of vibration and when the flashlight is shaken in the hand.

E–4h. Prefocused types shall be so constructed that they require no adjusting mechanism for focusing the light beam.

E–4i. The head of flashlights styles RF and PRF shall be either fixed or adjustable with respect to the body tube. If fixed it shall be so firmly fastened as not to be dislodged in ordinary use.

E–4j. Flashlights of styles RF and PRF shall be provided with a substantial belt hanger so mounted that the beam may be projected horizontally forward when the flashlight is hung on waist belt.

E–5. *Exterior finish.*—

E–5a. The exterior finish shall be as specified in requests for bids or orders, as follows:

1. Type I: As per paragraph B–1e (1).
2. Type II: As per paragraph B–1e (2).
3. Other finishes as may be specified.

E–5b. Chromium plating on external parts shall be applied to brass over a prior plating of nickel in such a manner as to be thoroughly adherent. Plated parts shall be bright polished and free of stains, dull spots, burns, or roughness.

E–5c. Black enamel shall be applied so as to cover the coated parts completely and shall be baked on. It shall be hard and adherent.

E–6. *Lamp holder.*—All flashlights, excepting prefocused styles, shall be fitted with a lamp receptacle adapted to receive a miniature screw base lamp.

E–6a. Prefocused styles (PF–2–D, PF–3–D, PS–3–D, PRF–2–D, PFI–2–D and PFI–3–D) shall have the lamp so mounted as to be held firmly in position with the flange of the lamp base received in a recess in the back of the reflector, and so located thereby that the filament is brought to the focal point of the reflector with the accuracy necessary to produce a spot of projected light as specified in paragraph F–3b. The recess at the back of the reflector shall be adapted to receive snugly the flanged base of the prefocused lamps, such as trade numbers PR–2 and PR–3. Means shall be provided for conveniently removing and replacing lamps from the back of the reflector.

E–6b. All focusing and prefocused flashlights shall have shock-absorbing means for protecting the lamp from breakage by battery impact.

E–7. *Marking.*—Each flashlight shall be marked with the manufacturer's name or trademark.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. *Inspection at place of manufacture.*—If inspection at the place of manufacture is made, it shall consist of the following:

F–1a. Not less than 6 nor more than 25 samples from each type, style, and grade shall be inspected to ascertain that the dimensions, the character of workmanship, and the material used are in conformity with requirements of this specification; and also a general inspection covering the operation of the switch and flasher, contact springs, assembly and disassembly of the reflector, lens, and lamp.

F–1b. Not less than six sample flashlights shall be selected at random from each lot submitted for inspection, fitted with batteries, and operated in the dark to observe the concentration and distribution of light throughout the beam.

F–1c. The flashlights selected for the above tests shall also be tested to ascertain that the case is nonshort-circuiting, that is, that the battery circuit cannot be closed by metallic contact outside the case.

F–2. *Laboratory tests.*—If laboratory tests are required, not less than six samples of each type and grade shall be furnished to the testing laboratory. Two samples shall be tested for each item, F–2a and F–2b, and F–3a, F–3b, or F–3c.

F–2a. The switch mechanism shall be operated in a normal manner through 25,000 operating cycles at a rate not to exceed 15 cycles per minute. Mechanical faults, contact, or other failures shall be reported as grounds for rejection.

F–2b. Reflection factor tests on reflectors shall be made with the proper lamp as specified in paragraphs D–2 and D–3.

F–3. *Light projection characteristics.*—

F–3a. Light projected by a flashlight having an N or NI style head shall show on a screen normal to the axis of the projected light and at a distance of 1 foot from the lens a circle not less than 10 inches in diameter nor greater than 18 inches in diameter. Light over this circle shall be essentially free of dark streaks and the illumination shall be approximately uniform throughout the illuminated spot except that a somewhat brighter center is permissible and desirable.

F–3b. The light projected by a flashlight having an F, FI, PF, RF, PRF, or PFI style head shall show on a screen normal to the axis of the projected light and at a distance of 5 feet from the lens a spot of concentrated light not greater than 9 inches in diameter.

F–3c. The light projected by a flashlight having an S-type head shall show on a screen at a distance of 5 feet and normal to the axis of the projected light a spot of concentrated light not greater than 10 inches in diameter.

F–4. *Inflammability:* Cut a sample of the insulating material from the side of the case approximately ½ inch wide and 5 inches long. Lines shall be drawn across the specimen at ½-inch intervals measured along the surface and starting from one end. Clamp the other end in a rigid support so that the longitudinal axis of the specimen is horizontal and the transverse axis is inclined at 45° to the horizontal. An alcohol lamp or gas burner with a flame ½ to ¾ inch in height shall be placed under the free end of the specimen so adjusted that the flame tip is just in contact with the material under test. A stop watch shall be started when the flame is placed in contact with the specimen. At the end of 10 seconds the flame shall be removed and the specimen allowed to burn. The rate of burning shall not exceed 3 inches per minute.

F–5. *Retests.*—If failure occurs in any item or items of paragraphs F–1, F–2, F–3, or F–4, an additional set of samples may be tested at the request of the purchaser. If a failure shall occur in the retest, this fact shall be reported as ground for rejection of the lot represented by the two sets of samples.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT.

G–1. *Packaging.*—Unless otherwise specified in requests for bids, commercial packages will be acceptable, under this specification.

G–2. *Packing.*—Unless otherwise specified, the flashlights shall be delivered in standard commercial containers, so constructed as to insure acceptance by common or other carrier, for safe transportation, at the lowest rate, to the point of delivery.

G-3. *Marking.*—Unless otherwise specified, shipping containers shall be marked with the name of the material, the type and quantity contained therein, as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order.

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS.

H-1. The following departmental specifications of the issue in effect on date of invitation for bids shall form a part of this specification.

H-1a. *Army.*—United States Army Specifications No. 100-2B, Standard Specification for Marking Shipments.

H-1a (1). The exterior finish of flashlights furnished the United States Army shall be as follows:

a. In the case of type I flashlights, olive-drab baked-on enamel shall be applied to the tube or body. End fittings, switch and belt hanger shall be oxidized nickel plated. This plating shall be adherent and durable and shall have a reflection factor no greater than that of the olive-drab enamel on the body.

b. In the case of type II flashlights, the exterior finish shall be of olive drab in color. The switch and end fittings shall be in olive drab or black.

H-1b. Navy Department purchase of flashlights will be made under the issue in effect on date of invitation for bids of Navy Department Specification 17F7, flashlights, electric, hand.

H-1c. *Panama Canal.*—Zinc-base die-cast metal reflectors will not be accepted by the Panama Canal.

I. NOTES.

I-1. Purchasers should exercise any desired options offered herein and should specify type, style, finish, etc.

I-2. It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

I-3. This specification covers only the types, styles, grades, sizes, etc., of the commodity as generally purchased by the Federal Government and is not intended to include all of the types, etc., which are commercially available.

I-4. An index of Federal specifications may be purchased as noted in paragraph next below, price to be obtained from the Superintendent of Documents.

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I-5. Copies of this specification and of W-B-101, and W-L-111, may be obtained upon application, accompanied by money order or coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington, D. C. Price 5 cents each.

Notice.—When Government drawings, specifications, or other data, are used for any purpos other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in anyway be related thereto.